

Thomas Ireland

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EDUCATION

University of Alberta

BSc. Engineering Physics conc. Nanoengineering

Months available

GPA

Starting Month

Edmonton, AB

Sept. 2024 – May 2029

4 – 12 months

3.3/4.0

January 2027

WORK EXPERIENCE

Python Coding Camp Director

Upper Canada College Summer Camps

July 2022 – Aug. 2025

Toronto, ON

- Promoted to Director from camp counsellor in June 2024
- Developed class materials, created weekly schedules, ran coding-based competitions
- Taught classes of 20 children aged 9-16 introductory Python, updating the teaching style for each student based on their level of understanding
- Managed classroom by organizing and maintaining the program's physical equipment, along with digital records

PROJECTS

Computer Integration Board | PCB Design, Soldering, CAN Bus, PCB Testing

Feb. 2026 – March 2026

- Designed a PCB to convert computer Data into CAN integrating the central computer with the CAN bus

Password Manager | Git, Python, cryptography, pandas, SQLite

June 2026 – July 2026

- Developed a locally ran password manager storing data using a SQLite database
- Implemented a Command-line tool which showed passwords cleanly

Barbecue Thermometer | PCB Design, C/C++, HTML/CSS, ESP32, I²C

July 2026 – Present

- Developing a ESP32-based thermometer which sends temperature data to my phone
- Designed a custom ESP32 dev-board, USB-C LiPo battery charging, and Battery monitoring Using Altium
- Building Firmware & website which processed the data and displayed it in an elegant way

Buoyancy Brace | SolidWorks

January 2025 – March 2025

- Designed a 3D printed brace for the buoyancy pods used on a robotic submarine

STUDENT ORGANIZATIONS

Electrical Team Member

Autonomous Robotic Vehicle Project

Sept. 2024 – Present

Edmonton, AB

- Built the Computer Integration Board and Buoyancy Brace
- Collaborated with a multidisciplinary team to design, assess, and improve electrical and mechanical components for an autonomous submarine
- Communicated with co-leads during team meetings updating them on project progress and setting deadlines

VP Social

Engineering Physics Club

Sept. 2026 – May 2027

Edmonton, AB

- Planning and organizing club social events, working within the club as well as with other engineering clubs
- Run social media accounts to inform members of upcoming events by using graphic design tools like Canva to make interesting posts

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (SQLite), HTML/CSS, Java

Hardware: Altium (Schematic & PCB design), Soldering, Crimping, PCB Testing, SolidWorks

Software: Git, Linux VS code, ESP-IDF

Libraries: pandas, NumPy, Matplotlib, SQLite3, cryptography